Largest Concatenation-Component in Regular Expressions

Hyperscan implements algorithms to decompose regexes into literals and FA in its Rose engine. There may be a relation between how well a regex can be decomposed and the size of its largest concatenation-component.

1.1 Recurrences

Let $\mathcal{A}$ be a finite alphabet and $\mathcal{T}$ be the set of all regexes excluding ϵ over $\mathcal{A}$. The size function $|\cdot|$ is the number of nodes.

Definition 1 (BST-like distribution for regexes)**.** *We define $p : \mathcal{T} \to [0, 1]$ as follows.*

1. $\sum_{c\in\mathcal{A}} p(c) = 1,$

2. $p\left(\begin{array}{c}\bullet\\ / \ \backslash \\ T_1 \quad T_2\end{array}\right) = \dfrac{u}{n-2}p(T_1)p(T_2),$ *where* $|T_1| + |T_2| = n - 1.$

3. $p\left(\begin{array}{c}+\\ / \ \backslash \\ T_1 \quad T_2\end{array}\right) = \dfrac{v}{n-2}p(T_1)p(T_2),$ *where* $|T_1| + |T_2| = n - 1.$

4. $p\left(\begin{array}{c}*\\ | \\ T\end{array}\right) = (1 - u - v)p(T).$

We want to compute the expected size of the largest all-concatenation component in a regex over all of equivalent forms of its parse tree. The reason is as follows. Take the regex

$$(abc + d)(xy + z) = abcxy + abcxz + dxy + dz.$$

We expect the largest concatenation-component to be in $abcxy$ instead of abc because the former is extracted by Hyperscan. Therefore, a subtree may *yield* its concatenation-component to its parent if the parent is a concatenation node.

Recall the construction

$$\mathcal{T} = \sum_{c\in\mathcal{A}} c + \begin{array}{c}*\\ | \\ \mathcal{T}\end{array} + \begin{array}{c}\bullet\\ / \ \backslash \\ \mathcal{T} \quad \mathcal{T}\end{array} + \begin{array}{c}+\\ / \ \backslash \\ \mathcal{T} \quad \mathcal{T}\end{array}.$$

Define the random variables

1. M_n: the largest concatenation-component that is not inside a Kleene star of a regex of size n *itself*,

2. Y_n: the largest concatenation-component that is not inside a Kleene star of a regex of size n that it *yields* to its parent.

We have the following recurrences

$$Y_{n+2} = \begin{cases} Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1 & \text{prob. } u, \\ \max(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}) & \text{prob. } v, \\ 0 & \text{prob. } 1-u-v \end{cases} \qquad Y_1 = Y_2 = 0. \quad (1.1)$$

$$M_{n+2} = \begin{cases} \max(Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1, M^{(1)}_{U_n}, M^{(2)}_{n+1-U_n}) & \text{prob. } u, \\ \max(M^{(1)}_{U_n}, M^{(2)}_{n+1-U_n}) & \text{prob. } v \\ 0 & \text{prob. } 1-u-v \end{cases} \qquad M_1 = M_2 = 0. \quad (1.2)$$

We have used $Y^{(1)}_n, Y^{(2)}_n, M^{(1)}_n, M^{(2)}_n$ as independent copies of Y_n and M_n respectively, and U_n as a uniform random variable over $\{1, 2, \dots, n\}$ independent of all $Y^{(i)}_n$ and $M^{(i)}_n$.

1.2 The yield recurrence

We rewrite the recurrence (1.1) as

$$Y_{n+2} \overset{d}{=} I_n \left(Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1 \right) + J_n \max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n} \right). \tag{1.3}$$

In (1.3), (I_n, J_n) is a pair of mutually exclusive Bernoulli random variables with $\mathbb{E}[I_n] = u$ and $\mathbb{E}[J_n] = v$ such that $I_n + J_n \le 1$. The sequences $\boldsymbol{Y}^{(1)} = (Y^{(1)}_n)_{n\ge1}$ and $\boldsymbol{Y}^{(2)} = (Y^{(2)}_n)_{n\ge1}$ are independent copies of the sequence $\boldsymbol{Y} = (Y_n)_{n\ge1}$. The notion of equality is in distribution. Note that we can reuse the same variables $Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}$ and U_n across both terms because at most one indicator is non-zero for any given realization. Let $\boldsymbol{I} = (I_n)_{n\ge1}$, and define $\boldsymbol{J}$ and $\boldsymbol{U}$ similarly. We assume that the sequence of pairs $(\boldsymbol{I}, \boldsymbol{J})$ and the sequence $\boldsymbol{U}$ each consist of independent and identically distributed elements. Furthermore, the entire stochastic processes $(\boldsymbol{I}, \boldsymbol{J})$, $\boldsymbol{U}$, $\boldsymbol{Y}^{(1)}$, and $\boldsymbol{Y}^{(2)}$ are mutually independent.

1.2.1 Lower and Upper Bounds of the Expectation

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^{n} y_i + \frac{v}{n} \sum_{i=1}^{n} \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, we use the fact that $\max(Y_i, Y_{n+1-i}) \le Y_i + Y_{n+1-i}$, which gives an upper bound sequence

$$y^U_{n+2} = u + \frac{2(u+v)}{n} \sum_{i=1}^{n} y^U_i, \quad y^U_1 = y^U_2 = 0. \tag{1.4}$$

For a lower bound, we use

$$\begin{aligned} \mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E}\left[\frac{1}{2}(Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\ &\ge \frac{1}{2}(\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2}|\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\ &= \frac{1}{2}(y_i + y_{n+1-i} + |y_i - y_{n+1-i}|). \end{aligned}$$

This gives the lower bound sequence

$$y^L_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^{n} y^L_i + \frac{v}{2n} \sum_{i=1}^{n} |y^L_i - y^L_{n+1-i}|, \quad y^L_1 = y^L_2 = 0. \tag{1.5}$$

Remark. Another lower bound is obtained as follows. By Jensen's inequality

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i,n+1-i)},$$

where the last equality holds because y_n is non-decreasing in n *(we have to prove it)*. This gives the following lower bound sequence

$$y^L_{n+2} = u + \frac{2u}{n}\sum_{i=1}^{n} y^L_i + \frac{2v}{n}\sum_{i=\lfloor (n+1)/2 \rfloor}^{n} y^L_i, \qquad y^L_1 = y^L_2 = 0. \tag{1.6}$$

Looking at the bounds (1.5), we know that when $2u + v > 1$, (y_n) grows at least polynomially.

1.2.2 Asymptotic of the Expectation

Consider the case $2u + v > 1$. By the bounds, there exists

$$\alpha = \inf\left\{\beta : \frac{Y_n}{n^\beta} \xrightarrow{d} 0\right\}.$$

We know that $\alpha \in (2u + v - 1, 2(u + v) - 1]$. We want to see if indeed $y_n \sim Cn^\alpha$ for some $C > 0$.

We try the contraction method [2, 3, 1]. Let us mention some conventions. For a random variable X, denote by $\mathcal{L}(X)$ its distribution. Let $(\mathcal{M}, d)$ be a metric space where $\mathcal{M}$ is a set of probability distributions of our random variables and the metric d is such that $d(\mathcal{L}(X), \mathcal{L}(Y)) = 0$ implies $X \overset{d}{=} Y$. The idea is as follows

1. Guess a normalization X_n of Y_n that converges in distribution.
2. Plug X_n into the recurrence. Assume that X_n converges, show that the limit equation has a unique solution X^*. This is done by proving that the operator T defined by the right-hand side of the limit equation is a contraction.
3. Prove that indeed $d(\mathcal{L}(X_n), \mathcal{L}(X^*)) \to 0$.

The method is use to obtain more information about the limit distribution of X_n rather than just its expectation. For example, the following is the recurrence for the number of comparisons in Quicksort where the first element is chosen as the pivot [1]

$$Y_n = Y^{(1)}_{U_n} + Y^{(2)}_{n-1-U_n} + n - 1.$$

One can find the asymptotic of $\mathbb{E}[Y_n]$ by taking expectation on both sides. The contraction is then used to show that $X_n = \dfrac{Y_n - \mathbb{E}[Y_n]}{n} \to X^*$, where the denominator n is the "guessed" variance.

Our case is weaker. The normalization is $X_n = Y_n/n^\alpha$ such that $X_n \xrightarrow{d} X^*$. Plugging into the recurrence (1.3), we have

$$\begin{aligned}(n+2)^\alpha X_{n+2} \overset{d}{=} I\left(U_n^\alpha X^{(1)}_{U_n} + (n+1-U_n)^\alpha X^{(2)}_{n+1-U_n} + 1\right) \\ + J\max\left(U_n^\alpha X^{(1)}_{U_n}, (n+1-U_n)^\alpha X^{(2)}_{n+1-U_n}\right)\end{aligned}$$

Equivalently,

$$\begin{aligned}X_{n+2} \overset{d}{=} I\left(\left(\frac{U_n}{n+2}\right)^\alpha X^{(1)}_{U_n} + \left(\frac{n+1-U_n}{n+2}\right)^\alpha X^{(2)}_{n+1-U_n} + \frac{1}{(n+2)^\alpha}\right) \\ + J\max\left(\left(\frac{U_n}{n+2}\right)^\alpha X^{(1)}_{U_n}, \left(\frac{n+1-U_n}{n+2}\right)^\alpha X^{(2)}_{n+1-U_n}\right)\end{aligned} \tag{1.7}$$

Proposition 1. *The right-hand side of the above recurrence converges in distribution to*

$$I\left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}\right) + J \max\left(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}\right) \tag{1.8}$$

Proof. We first show that

$$\left(\frac{U_n}{n+2}\right)^\alpha X^{(1)}_{U_n} + \left(\frac{n+1-U_n}{n+2}\right)^\alpha X^{(2)}_{n+1-U_n} \xrightarrow{d} U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}.$$

By Lemma **??** and Slutsky's theorem,

$$\frac{U_n}{n+2} = \frac{U_n}{n} \cdot \frac{n}{n+2} \xrightarrow{d} U.$$

To apply the Continuous Mapping Theorem, we need to show that

$$\left(\frac{U_n}{n+2}, X^{(1)}_{U_n}, X^{(2)}_{n+1-U_n}\right) \xrightarrow{d} \left(U, X^{(1)}, X^{(2)}\right).$$

Since $U, X^{(1)}, X^{(2)}$ are independent, this is equivalent to showing that $\forall u, x_1, x_2 \in \mathbb{R}$

$$\mathbb{P}\left(U_n \le \lfloor nu \rfloor, X^{(1)}_{U_n} \le x_1, X^{(2)}_{n+1-U_n} \le x_2\right) \longrightarrow \mathbb{P}(U \le u)\mathbb{P}(X^{(1)} \le x_1)\mathbb{P}(X^{(2)} \le x_2).$$

Indeed,

$$\begin{aligned}
&\mathbb{P}\left(U_n \le \lfloor nu \rfloor, X^{(1)}_{U_n} \le x_1, X^{(2)}_{n+1-U_n} \le x_2\right) \\
&= \sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P}(U_n = k)\mathbb{P}\left(X^{(1)}_k \le x_1, X^{(2)}_{n+1-k} \le x_2 | U_n = k\right) \\
&= \frac{1}{n}\sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P}\left(X^{(1)}_k \le x_1, X^{(2)}_{n+1-k} \le x_2\right) && (U_n \perp (X^{(1)}_k, X^{(2)}_{n+1-k})) \\
&= \frac{1}{n}\sum_{k=1}^{\lfloor nu \rfloor} \mathbb{P}\left(X^{(1)}_k \le x_1\right)\mathbb{P}\left(X^{(2)}_{n+1-k} \le x_2\right) && (X^{(1)}_k \perp X^{(2)}_{n+1-k})
\end{aligned}$$

Let F_n be the common CDF of $X^{(1)}_n$ and $X^{(2)}_n$, and F be the CDF of $X^{(1)}$. Let $m = \lfloor nu \rfloor$. The problem becomes showing that

$$\frac{1}{n}\sum_{k=1}^{m} F_k(x_1)F_{n+1-k}(x_2) \longrightarrow uF(x_1)F(x_2).$$

Equivalently,

$$\frac{m}{n}\left(\frac{1}{m}\sum_{k=1}^{m}\left(F_k(x_1)F_{n+1-k}(x_2) - F(x_1)F(x_2)\right)\right) \longrightarrow 0.$$

We have

$$
\begin{aligned}
&\frac{1}{m}\sum_{k=1}^{m}\left|F_k(x_1)F_{n+1-k}(x_2)-F(x_1)F(x_2)\right| \\
&\le \frac{1}{m}\sum_{k=1}^{m}\left|F_k(x_1)\left(F_{n+1-k}(x_2)-F(x_2)\right)\right| + \frac{1}{m}\sum_{k=1}^{m}\left|F(x_2)\left(F_k(x_1)-F(x_1)\right)\right| \\
&\le \frac{1}{m}\sum_{k=1}^{m}\left|F_{n+1-k}(x_2)-F(x_2)\right| + \frac{1}{m}\sum_{k=1}^{m}\left|F_k(x_1)-F(x_1)\right| \qquad (|F_n|\le 1)
\end{aligned}
$$

The second term goes to zero by the fact that $F_n(x_1)\to F(x_1)$ and Cesàro Means. We can write the first term as

$$
\frac{1}{m}\sum_{k=n-m+1}^{n}\left|F_k(x_2)-F(x_2)\right|.
$$

If $u=1$, its convergence is similar to the second term. If $u<1$, then $n-m+1\to\infty$. By pointwise convergence,$\forall\epsilon>0,\exists N,\forall n\ge N:|F_n(x_2)-F(x_2)|<\epsilon$. Hence, for $n-m+1\ge N$,

$$
\frac{1}{m}\sum_{k=n-m+1}^{n}\left|F_k(x_2)-F(x_2)\right|<\epsilon.
$$

Using Slutsky's theorem again, we have

$$
\left(\frac{n+1}{n+2},\frac{U_n}{n+2},X^{(1)}_{U_n},X^{(2)}_{n+1-U_n}\right)\xrightarrow{d}\left(1,U,X^{(1)},X^{(2)}\right).
$$

Now define the function $f:\mathbb{R}_+^4\cap\{(t,u,x,y):t\ge u\}\to\mathbb{R}$ by

$$
f(t,u,x,y)=u^\alpha x+(t-u)^\alpha y.
$$

Then f is continuous. The Continuous Mapping Theorem applies. The convergence

$$
\max\left(\left(\frac{U_n}{n+2}\right)^\alpha X^{(1)}_{U_n},\left(\frac{n+1-U_n}{n+2}\right)^\alpha X^{(2)}_{n+1-U_n}\right)\xrightarrow{d}\max\left(U^\alpha X^{(1)},(1-U)^\alpha X^{(2)}\right)
$$

is similar. And finally, we define the continuous function $g(i,j,u,v)=iu+jv$ on $\mathbb{R}^4$ to finish the proof.

□

Proposition 2. *Let $T:\mathcal{M}\to\mathcal{M}$ be an operator in the space of probability distributions with finite expectation, defined by*

$$
T(\mu)=\mathcal{L}\left(I(U^\alpha X^{(1)}+(1-U)^\alpha X^{(2)})+J\max(U^\alpha X^{(1)},(1-U)^\alpha X^{(2)})\right),
$$

where $X^{(1)},X^{(2)}\overset{iid}{\sim}\mu$, U is uniformly distributed on $[0,1]$, I and J are Bernoulli with expectation u and v respectively, such that $I+J\le 1$. The variables U,I,J and $(X^{(1)},X^{(2)})$ are mutually independent. Then there exists $p\ge 2$ such that T is a contraction in the $(\mathcal{M}_p,\ell_p)$, where the ℓ_p metric is defined by

$$
\ell_p(\mu,\nu)=\inf\{\mathbb{E}[|X-Y|^p]:X\sim\mu,Y\sim\nu\}.
$$

Proof. Let $\mu, \nu \in \mathcal{M}_p$. Let $X, X^{(1)}, X^{(2)} \overset{\text{iid}}{\sim} \mu$ and $Y, Y^{(1)}, Y^{(2)} \overset{\text{iid}}{\sim} \nu$ be independent of each other. We have

$$\begin{aligned}
\ell_p(T(\mu), T(\nu)) = \mathbb{E}\Big[\Big| & I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) \\
& - I(U^\alpha Y^{(1)} + (1-U)^\alpha Y^{(2)}) - J\max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)})\Big|\Big] \\
\le \mathbb{E}\Big[\Big| & I\left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)}\right)\Big|\Big] \\
& + \mathbb{E}\left[\left|J\left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)})\right)\right|\right]
\end{aligned}$$

Consider the first term. We have

$$\begin{aligned}
& \mathbb{E}\left[\left|I\left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)}\right)\right|\right] \\
= & \mathbb{E}\left[I\left|\left(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)} - U^\alpha Y^{(1)} - (1-U)^\alpha Y^{(2)}\right)\right|\right] && (I \ge 0) \\
= & u\mathbb{E}\left[\left|U^\alpha\left(X^{(1)} - Y^{(1)}\right) + (1-U)^\alpha\left(X^{(2)} - Y^{(2)}\right)\right|\right] && (\text{independence of } I) \\
\le & u\left(\mathbb{E}\left[U^\alpha\left|X^{(1)} - Y^{(1)}\right|\right] + \mathbb{E}\left[(1-U)^\alpha\left|X^{(2)} - Y^{(2)}\right|\right]\right) && (U, 1-U \ge 0) \\
= & \frac{2u}{\alpha+1}\mathbb{E}[|X - Y|].
\end{aligned}$$

We turn to the second term. Recall that $|\max(a,b) - \max(c,d)| \le \max(|a-c|, |b-d|)$ and $\max(a,b) = \frac{1}{2}(a + b + |a-b|)$ for any real numbers a, b, c, d. We have

$$\begin{aligned}
& \mathbb{E}\left[\left|J\left(\max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) - \max(U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)})\right)\right|\right] \\
\le & v\mathbb{E}\left[\max\left(\left|U^\alpha\left(X^{(1)} - Y^{(1)}\right)\right|, \left|(1-U)^\alpha\left(X^{(2)} - Y^{(2)}\right)\right|\right)\right] \\
= & \frac{v}{2}\Big(\mathbb{E}\left[U^\alpha\right]\mathbb{E}\left[\left|X^{(1)} - Y^{(1)}\right|\right] + \mathbb{E}\left[(1-U)^\alpha\right]\mathbb{E}\left[\left|X^{(2)} - Y^{(2)}\right|\right] \\
& + \mathbb{E}\left[\left|U^\alpha\left(X^{(1)} - Y^{(1)}\right) - (1-U)^\alpha\left(X^{(2)} - Y^{(2)}\right)\right|\right]\Big) \\
= & \frac{v}{\alpha+1}\mathbb{E}\left[|X - Y|\right] + \frac{v}{2}\mathbb{E}\left[\left|U^\alpha\left(X^{(1)} - Y^{(1)}\right) - (1-U)^\alpha\left(X^{(2)} - Y^{(2)}\right)\right|\right]
\end{aligned}$$

Combining the bounds for both terms, we get

$$\ell_1(T(\mu), T(\nu)) \le \frac{2u}{\alpha+1}\mathbb{E}[|X - Y|] + J\mathbb{E}[|X - Y|].$$

Since $\dfrac{2u}{\alpha+1} + J < 1$, the operator T is a contraction in the $(\mathcal{M}_1, \ell_1)$ space.

$\square$

Asymptotic Analysis of Some Sequences

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In this section, we analyze the asymptotic behavior of lower bounds of $\mathbb{E}[Y_n]$, as follows.

$$x_{n+2} = \frac{a}{n}\sum_{i=1}^{n} x_i + b, \quad n \geq 1, \quad x_1 = x_2 = 0. \tag{2.1}$$

$$x_{n+2} = \frac{a}{n}\sum_{i=1}^{n} x_i + \frac{b}{n}\sum_{i=1}^{n} |x_i - x_{n+1-i}| + c, \quad n \geq 1, \quad x_1 = x_2 = 0 \quad (a > 1, 0 < b < 1, c > 0) \tag{2.2}$$

$$x_n = \frac{2u}{n}\sum_{i=0}^{n-1} x_i + \frac{2v}{n}\sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} x_i, x_0 = x_1 = 1 \quad (u, v > 0, u + v \leq 1, 2u + v > 1). \tag{2.3}$$

Proposition 3. *Consider recurrence* (2.1). *We have*

- *If* $a < 1$, *then* $x_n \sim \dfrac{b}{1-a}$.
- *If* $a = 1$, *then* $x_n \sim b \ln n$.
- *If* $a > 1$, *then* $x_n \sim \dfrac{b}{(a-1)\Gamma(a)} n^{a-1}$.

Proof. Consider the generating function $X(z) = \sum_{n=1}^{\infty} x_n z^n$ with the convention that $x_0 = 0$. We rewrite the recurrence as

$$(n+2)x_{n+2} - 2x_{n+2} = a\sum_{i=1}^{n} x_i + bn.$$

We multiply both sides by z^{n+2} and sum over $n \geq 1$. The left-hand side becomes

$$\begin{aligned}
\sum_{n=1}^{\infty}((n+2)x_{n+2} - 2x_{n+2})z^{n+2} &= \sum_{n=3}^{\infty} n x_n z^n - 2\sum_{n=3}^{\infty} x_n z^n \\
&= z\sum_{n=1}^{\infty} n x_n z^{n-1} - 2\sum_{n=1}^{\infty} x_n z^n \\
&= zX'(z) - 2X(z).
\end{aligned}$$

The right-hand side becomes

$$\begin{aligned}
\sum_{n=1}^{\infty}(a\sum_{i=1}^{n} x_i + bn)z^{n+2} &= az^2\sum_{n=1}^{\infty}\sum_{i=1}^{n} x_i z^n + bz^2\sum_{n=1}^{\infty} n z^n \\
&= az^2\frac{X(z)}{1-z} + \frac{bz^3}{(1-z)^2}.
\end{aligned}$$

Therefore, the generating function $X(z)$ satisfies the following ODE

$$X'(z) - \left(\frac{2}{z} + \frac{az}{1-z}\right) X(z) = \frac{bz^3}{(1-z)^2}.$$

We have $\mu(z) = \exp\left(\int -\left(\frac{2}{z} + \frac{a}{1-z} - a\right) \mathrm{d}z\right) = z^{-2}(1-z)^a$. which gives

$$z^{-2}(1-z)^a X(z) = \int b(1-z)^{a-2} \, \mathrm{d}z.$$

There are two cases of a that determine the integral. If $a = 1$, we have

$$z^{-2}(1-z)X(z) = b \ln\left(\frac{1}{1-z}\right) + C.$$

Since $x_1 = x_2 = 0$, we have $z^{-2}X(z) \to 0$ as $z \to 0$, which gives $C = 0$. Therefore, we have

$$X(z) = b\frac{z^2}{1-z} \ln\left(\frac{1}{1-z}\right).$$

Hence, $x_n = bH_{n-2} \sim b \ln n$.

If $a \neq 1$, we have

$$z^{-2}(1-z)^a X(z) = -\frac{b}{a-1}(1-z)^{a-1} + C.$$

Using the same argument as above, we have $C = \dfrac{b}{a-1}$. Therefore, we have

$$X(z) = \frac{b}{a-1} z^2 \left((1-z)^{-a} - (1-z)^{-1}\right).$$

Using the generalized binomial theorem

$$(1-z)^{-a} = \sum_{n=0}^{\infty} \binom{n+r-1}{n} z^n,$$

we have

$$x_n = \frac{b}{a-1}\left(\binom{n-3+a}{n-2} - 1\right) \sim \frac{b}{a-1}\left(\frac{n^{a-1}}{\Gamma(a)} - 1\right).$$

If $a < 1$ then $x_n \sim \dfrac{b}{1-a}$. If $a > 1$, then $x_n \sim \dfrac{b}{(a-1)\Gamma(a)} n^{a-1}$. $\square$

Lemma 1. *The sequence (x_n) in (2.2) is increasing.*

Proof. One checks that $x_n \geq 0$ for all n. Let us prove that (x_n) is increasing. We have $x_2 \geq x_1$.

Assume that $x_n \geq x_{n-1} \geq \ldots \geq x_0$ for some $n \geq 3$. Then,

$$\begin{aligned}
nx_{n+2} &= a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} |x_i - x_{n+1-i}| + nc \\
&= a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{\lfloor n/2\rfloor} (x_{n+1-i} - x_i) + nc \\
&= a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{\lfloor n/2\rfloor} (x_{n+1-i} - x_i) + nc \\
&= a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{\lfloor n/2\rfloor} (x_{n+1-i} - x_i) + nc \\
&= a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{\lfloor n/2\rfloor} (x_{n+1-i} - x_i) + nc \\
&= (a+b)\sum_{i=1}^{n} x_i - 2b\sum_{i=1}^{\lfloor n/2\rfloor} x_i - bm_n + nc,
\end{aligned}$$

where

$$m_n = \begin{cases} x_{\lfloor n/2\rfloor+1} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases}$$

Let $S_n = \sum_{i=1}^{n} x_i$. Then,

$$\begin{aligned}
nx_{n+2} &= (a+b)S_n - 2bS_{\lfloor n/2\rfloor} - bm_n + nc, \\
S_{n+1} &= S_n + x_{n+1}.
\end{aligned}$$

Hence,

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2b(S_{\lfloor n/2\rfloor} - S_{\lfloor (n-1)/2\rfloor}) - b(m_n - m_{n-1}) + c.$$

Let us check the right-hand side according to the parity of n.

- If $n = 2k$, then $S_{\lfloor n/2\rfloor} - S_{\lfloor (n-1)/2\rfloor} = x_k$ and $m_n - m_{n-1} = -x_k$. The right-hand side is $(a+b)x_n - 2bx_k + c$.
- If $n = 2k+1$, then $S_{\lfloor n/2\rfloor} - S_{\lfloor (n-1)/2\rfloor} = 0$ and $m_n - m_{n-1} = x_{k+1}$. The right-hand side is $(a+b)x_n - 2bx_{k+1} + c$.

Collecting the two cases, we have

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lceil n/2\rceil} + c.$$

The proof follows from Proposition **??**. □

Lemma 2. *The sequence in* (2.2) *can be rewritten as*

$$nx_{n+2} - (n-1)x_{n+1} = (a+b)x_n - 2bx_{\lceil n/2\rceil} + c. \tag{2.4}$$

Proposition 4. *Consider the sequence defined by* (2.3). *We have $x_n = \Theta(n^{\alpha})$, where α is the unique solution of*

$$2(u+v) - v2^{-\alpha} = \alpha + 1.$$

Proof. For the lower bound, assume the inductive hypothesis $x_n \geq K(n^\alpha + 1)$ with $K \leq \dfrac{1}{2}$ to be determined. The base cases $n = 0, 1$ hold. Let us check for $n = 2$.

$$x_2 = 2u + 2v = \alpha + 1 + v2^{-\alpha} \geq \alpha + 1 \geq \frac{1}{2}(2^\alpha + 1), \forall \alpha \in [0, 1].$$

For $n \geq 3$, we have

$$\begin{aligned}
\frac{x_n}{K} &\geq \frac{2u}{n} \sum_{i=0}^{n-1} (i^\alpha + 1) + \frac{2v}{n} \sum_{i=\lfloor (n-1)/2 \rfloor}^{n-1} (i^\alpha + 1) \\
&\geq \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha \, \mathrm{d}x \right) + n \right) + \frac{2v}{n} \int_{(n-3)/2}^{n-1} (x^\alpha + 1) \, \mathrm{d}x \\
&\geq \frac{2u}{n} \left(\left(\int_0^{n-1} x^\alpha \, \mathrm{d}x \right) + n \right) + \frac{2v}{n} \left(\int_{(n-2)/2}^{n-1} x^\alpha + \frac{n+1}{2} \right) \mathrm{d}x \\
&= \frac{2u}{n} \left(\frac{(n-1)^{\alpha+1}}{\alpha + 1} + n \right) + \frac{2v}{n} \left(\frac{(n-1)^{\alpha+1} - (n/2)^{\alpha+1}}{\alpha + 1} + \frac{n+1}{2} \right) \\
&= \frac{(n-1)^{\alpha+1}}{n(\alpha+1)} \left(2u + 2v - v2^{-\alpha} \right) + 2u + v \left(1 + \frac{1}{n} \right) \\
&\geq n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v
\end{aligned}$$

We are done if we can prove that for every $n \geq 3$,

$$n^\alpha \left(1 - \frac{1}{n} \right)^{\alpha+1} + 2u + v \geq n^\alpha + 1.$$

By Bernoulli's inequality, we have $\left(1 - \dfrac{1}{n} \right)^{\alpha+1} \geq 1 - \dfrac{\alpha+1}{n}$. Hence, it suffices to show that

$$n^\alpha \left(1 - \frac{\alpha + 1}{n} \right) + 2u + v \geq n^\alpha + 1 \iff 2u + v - 1 \geq \frac{\alpha + 1}{n^{1-\alpha}}.$$

Let $N_0 = \left\lceil \left(\dfrac{\alpha + 1}{2u + v - 1} \right)^{\frac{1}{1-\alpha}} \right\rceil$. The inequality holds for every $n \geq N_0$. Now we choose K for it to hold for $n < N_0$. In particular,

$$K = \min \left(\frac{1}{2}, \min_{3 \leq i < N_0} \frac{x_i}{i^\alpha + 1} \right) > 0.$$

The lower bound follows.

$\square$

Monte-Carlo Simulation for Checking Conjectures

May 04, 2026

We want to check the conjecture that the stochastic fixed-point equation

$$X \overset{d}{=} I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}) \tag{3.1}$$

has a unique solution (α^*, X_{α^*}) such that $\mathbb{E}[X_{\alpha^*}] = 1$. Let T_α be the operator defined by the right-hand side of the equation, i.e.,

$$T_\alpha(\mu) = I(U^\alpha X^{(1)} + (1-U)^\alpha X^{(2)}) + J \max(U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)}), \quad X^{(1)}, X^{(2)} \overset{\text{iid}}{\sim} \mu. \tag{3.2}$$

The classical contraction argument does not work. Since $X = 0$ is always a solution, we cannot use the Wasserstein metric W_p. Since T_α can change the mean of the distribution, we cannot use the Zolotarev metric ζ_p.

3.1 Background

3.1.1 Empirical Distributions

Given a distribution μ and the number of particles n, we can sample

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \mu.$$

The empirical distribution μ_n is

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}.$$

Definition 2 (Weak convergence of distributions)**.** *Let (M, d) be a metric space. Let $\mathcal{M}$ be the Borel σ-algebra on M. We define the distribution on $(M, \mathcal{M})$. Let $C_b(M)$ be the set of real-valued, bounded, continuous functions on M.*

A sequence of distributions (μ_n) converges weakly to a distribution μ if for $f \in C_b(M)$, we have

$$\int f \, d\mu_n \to \int f \, d\mu.$$

Theorem 1 (Varadarajan)**.** *The sequence μ_n converges weakly to μ almost surely.*

Remark. We need to study the rate of convergence, for simulation efficiency.

By Continuous Mapping Theorem, we can do arithmetic operations on the empirical distributions.

Proposition 5. *Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} \mu$ and $\mu_N = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$. Let $f : M \to M$ be a continuous function. Then*

$$\frac{1}{n} \sum_{i=1}^n \delta_{f(X_i)} \xrightarrow{w} f_\# \mu$$

almost surely.

We use the following particular cases

1. When $f(x,y) = x + y$, then $\frac{1}{n}\sum_{i=1}^{n}\delta_{X_i+Y_i} \xrightarrow{w} \mathcal{L}(X+Y)$ almost surely.
2. When $f(x,y) = \max(x,y)$, then $\frac{1}{n}\sum_{i=1}^{n}\delta_{\max(X_i,Y_i)} \xrightarrow{w} \mathcal{L}(\max(X,Y))$ almost surely.
3. when $f(x,y) = xy$, then $\frac{1}{n}\sum_{i=1}^{n}\delta_{X_iY_i} \xrightarrow{w} \mathcal{L}(XY)$ almost surely.

Proposition 6. *Let $(\mu_n) beempiricalmeasures$ such that $(\mu_n) \xrightarrow{w} \mu$ almost surely and the p-th moment of μ is finite. Then*

$$W_p(\mu_n, \mu) \to 0$$

almost surely.

Using the triangle inequality, we can show the following convergence result.

Proposition 7 (Convergence of Wasserstein distance)**.** *Let $(\mu_n) \xrightarrow{w} \mu$ almost surely and $\nu_n \xrightarrow{w} \nu$ almost surely Then*

$$W_p(\mu_n, \nu_n) \to W_p(\mu, \nu)$$

almost surely.

Proposition 7 provides an $O(n \log n)$ algorithm to compute the empirical Wasserstein distance

$$W_p(\mu_n, \nu_n) = \frac{1}{n}\inf_{i,j}\left(\sum_{i=1}^{n}|X_i - Y_j|^p\right)^{1/p} = \frac{1}{n}\left(\sum_{i=1}^{n}|\hat{X}_i - \hat{Y}_i|^p\right)^{1/p}, \tag{3.3}$$

where $(\hat{X}_1, \ldots, \hat{X}_n)$ and $(\hat{Y}_1, \ldots, \hat{Y}_n)$ are the sorted samples from μ_n and ν_n, respectively.

3.1.2 Random Distributions

Our conjectures include convergence of distributions. We need a way to generate random distributions. In particular, we need to generate random distributions with mean 1. A straightforward way is to use a bank of parametric distributions, such as uniform, log-normal, and gamma distributions. The base distribution is chosen randomly from the bank, and the shape parameters are also randomized. Finally, we can normalize the distribution to have mean 1.

A problem with the last approach is expressivity. Let us study Dirichlet processes.

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